UNIVERSITÀ DI BOLOGNA



Master's Degree in Automation Engineering

Modeling and Simulation of Mechatronic Systems

UR5 ROBOT BOND GRAPH MODEL

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Academic year 25/26

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Chapter 1

The Link Sub-Model

1.1 Physical Description

Each link of the UR5 is modelled as a rigid body characterised by:

- a mass m,
- an inertia tensor $J \in \mathbb{R}^{3\times3}$ expressed in the link frame,
- the position of the centre of mass r_{Gp} with respect to the link's proximal joint frame origin.

Numerical values for the UR5, taken from [**?**], are listed in table 1.1.

Table 1.1: UR5 link masses, centres of gravity, and diagonal inertia components.

Link	m [kg]	$r_{Gp,x}$ [m]	$r_{Gp,y}$ [m]	$r_{Gp,z}$ [m]
1	3.700	0.0000	−0.02561	0.00193
2	8.393	0.2125	0.0000	0.11336
3	2.330	0.1500	0.0000	0.02650
4	1.1219	0.0000	−0.00180	0.01634
5	1.1219	0.0000	0.00180	0.01634
6	0.1879	0.0000	0.0000	−0.001159

1.2 Ports and parameters

1.3 Equations of Motion

The Newton–Euler equations for link i, written in the body frame attached to the link, are given by **????**. Denoting the generalised momentum as

$$p_{\text{lin}} = m\, v_G, \qquad h_{\text{ang}} = J\, \omega, \tag{1.1}$$

one obtains the state-space form that 20-sim integrates directly:

$$\dot{p}_{\text{lin}} = f^{\text{ext}} + m\,\omega \times v_G, \tag{1.2}$$
$$\dot{h}_{\text{ang}} = \tau^{\text{ext}} + \omega \times J\omega, \tag{1.3}$$
$$v_G = p_{\text{lin}}/m, \tag{1.4}$$
$$\omega = J^{-1} h_{\text{ang}}. \tag{1.5}$$

1.4 Bond-Graph Structure

The `Link` sub-model exposes two *power ports* of dimension six:

- `wrench_i`: the wrench applied at the proximal joint (input, integral causality).
- `wrench_ipp`: the wrench at the distal joint (input, integral causality).

It also receives a signal input `T_next`, the homogeneous transformation, used to orient gravity.

Internally the model contains:

1. Two **MTF** elements (`i_com` and `ipp_com`) that shift the reference from the joint origins to the centre of mass via `????`.
2. A **1-junction** that enforces the common-velocity constraint at the centre of mass and sums all efforts.
3. The **Body** I-element that integrates the Newton–Euler equations.
4. A **MSe** element that injects the gravitational wrench.

1.4.1 Gravity Source (MSe)

The gravitational force in the inertial frame is $f_g = [0,\, 0,\, -mg]^T$. To express it in the body frame of link i, the rotation matrix $R_i \in SO(3)$ extracted from `T_next` is used:

$$f_g^{\text{body}} = R_i^T \begin{pmatrix} 0 \\ 0 \\ -mg \end{pmatrix}, \qquad \tau_g^{\text{body}} = \mathbf{0}. \tag{1.6}$$

This wrench is injected at the 1-junction so that it acts on the body mass.

1.4.2 Body sub-model

1.4.3 MTF Equations

For a displacement vector $r_{Gp} \in \mathbb{R}^3$ from the joint origin to the CoM:

$$v_G = v_{\text{joint}} + \omega \times r_{Gp}, \tag{1.7}$$
$$\tau_{\text{joint}} = \tau_G + r_{Gp} \times f_G. \tag{1.8}$$

These equations are implemented as a modulated transformer in 20-sim, where the modulation matrix is $[r_{Gp}]_\times$.

1.4.4 Causality Assignment

Both power ports carry *fixed integral causality* (the effort is known, the flow is computed). This means:

- Efforts (forces/torques at the joints) are *given* by the connected joints.
- Velocities (flows) are *computed* by the body integrator.

The 1-junction has one element with *effort-out* causality (the Body I-element) and all others with *effort-in* causality (the two MTF connections and the MSe source). The *constraint*: MTF elements with fixed causality on both sides impose a *causality conflict* unless the effort-output side is consistently identified — this is resolved by stipulating that the Body is the unique element on the 1-junction that imposes effort.

FIGURE PLACEHOLDER – Link Bond-Graph Diagram

Insert here the bond-graph diagram of the `Link` *sub-model, showing the two MTF elements connecting the proximal and distal joint ports to the 1-junction, the Body I-element, and the MSe gravity source. Causality strokes should be clearly indicated on each half-bond.*

Chapter 2

The Joint Sub-Model

2.1 Physical Description

Each of the six joints of the UR5 is a revolute joint. In the bond-graph model, the `Joint` sub-model is responsible for:

1. Performing the DH frame transformation between the inbound and outbound twist ports.
2. Extracting the scalar joint velocity $\dot{q}_i$ and position q_i.
3. Receiving the scalar actuator torque τ_i and transmitting it to the mechanical chain.

2.2 Ports and Parameters

Table 2.1: `Joint` sub-model ports.

Port	Type	Description
`twist_i`	power in (fixed-out causality)	Twist of frame $i-1$
`twist_ipp`	power out (fixed-out causality)	Twist of frame i
`actuator`	power in (fixed-in causality)	Actuator scalar torque bond
`T_i`	signal out $[4 \times 4]$	DH homogeneous transformation H_i^{i-1}
`Q`	signal out $[2 \times 1]$	Joint state $[q_i,\ \dot{q}_i]^T$

DH parameters stored in the `Joint` sub-model are d_i, a_i, α_i, and the offset $\tilde{q}_i$. These are set for each of the six joints according to table 2.2.

Table 2.2: UR5 Denavit–Hartenberg parameters.

Joint	d_i [m]	a_i [m]	α_i [rad]	$\tilde{q}_i$ [rad]
Base (1)	0.089159	0.0	$\pi/2$	0
Shoulder(2)	0.0	-0.425	0	0
Elbow (3)	0.0	-0.39225	0	0
Wrist1 (4)	0.10915	0.0	$\pi/2$	0
Wrist2 (5)	0.09465	0.0	$-\pi/2$	0
Wrist3 (6)	0.0823	0.0	0	0

2.3 The MTF Element as a DH Transformer

The core of the `Joint` model is an MTF element that implements the DH transformation. Given the joint angle $\theta_i = q_i + \tilde{q}_i$ and the stored parameters, the rotation matrix R_i and translation P_i are computed as follows (matching **??**):

$$R_i = \begin{pmatrix} \cos\theta_i & -\sin\theta_i \cos\alpha_i & \sin\theta_i \sin\alpha_i \\ \sin\theta_i & \cos\theta_i \cos\alpha_i & -\cos\theta_i \sin\alpha_i \\ 0 & \sin\alpha_i & \cos\alpha_i \end{pmatrix}, \tag{2.1}$$

$$P_i = \begin{pmatrix} a_i \cos\theta_i \\ a_i \sin\theta_i \\ d_i \end{pmatrix}. \tag{2.2}$$

The MTF enforces power continuity by transforming twists and wrenches:

$$\tau_{\text{ipp}} = R_i^T \tau_i, \tag{2.3}$$

$$f_{\text{ipp}} = R_i^T f_i - R_i^T (P_i \times \tau_i), \tag{2.4}$$

$$\omega_i = R_i\, \omega_{\text{ipp}}, \tag{2.5}$$

$$v_i = P_i \times R_i\, \omega_{\text{ipp}} + R_i\, v_{\text{ipp}}. \tag{2.6}$$

2.4 Joint Kinematics and Constraint Elements

The joint variable q_i (position) and $\dot{q}_i$ (velocity) are extracted from the power flow:

- A **FlowSensor** reads the scalar rotational velocity $\dot{q}_i = \omega_z$ on the z-axis of frame $i-1$.
- An **Integrator** computes $q_i = \int \dot{q}_i\, dt$.

The joint imposes the revolute constraint: only rotation about z is free. This is achieved via:

- A **ZeroJunction** for the xy-components of angular velocity (all equal and constrained to zero).
- A **ZeroJunction** for the z-component (free to rotate, actuated).
- A **0-junction** and constraint element (`Vel_constr`, `Omega_constr`) for translational and rotational velocities, enforcing that the joint is purely revolute.

TODO: SPIEGARE MEGLIO CONSTRAINTS E REVOLUTE PAIR

2.5 Causality Assignment and Complexity

The joint sub-model presents the most complex causality situation in the entire robot model:

- The inbound port `twist_i` has *fixed effort-out* causality: the joint *imposes* force/torque on the previous link and *receives* velocity from it.
- The outbound port `twist_ipp` has *fixed effort-out* causality as well: the joint imposes force/torque on the next link.
- The MTF has a *causality constraint*: the two sides must have opposite causality (one effort-in, one effort-out).
- The constraint elements (`Vel_constr` for translational DoFs, `Omega_constr` for xy-rotation) set effort from flow via an algebraic constraint equation, implying *derivative causality* on the constrained directions.
- The `actuator` port has *fixed-in causality*: the motor effort is imposed externally, so the 0-junction for q_z has *one effort-in* element (the actuator).

The combination of fixed causality on both external power ports and the internal MTF constraint means that *no free integral causality* can be assigned to the internal inertia elements of the joint body; instead, these joints rely on the inertia of the connected links. This is the correct physical interpretation: the joint itself has zero inertia in this model.

FIGURE PLACEHOLDER – Joint Bond-Graph Diagram

Insert here the bond-graph diagram of the `Joint` *sub-model. Show: the two power ports* `twist_i` *and* `twist_ipp`*, the central MTF implementing the DH transformation, the PowerDemux/PowerMux elements decomposing the 6-D twist into translational and rotational components, the 0-junctions and constraint blocks enforcing the revolute constraint, the FlowSensor and integrator extracting* $\dot{q}_i$ *and* q_i*, and the actuator bond. Causality strokes must be clearly marked.*

Chapter 3

The Transformation Sub-Model

3.1 Role in the Kinematic Chain

The `Transformation` sub-model is a *signal-only* block (no power ports) that computes the cumulative homogeneous transformation from the base frame to frame i. It is inserted between consecutive joints to propagate the absolute frame position and orientation.

3.2 Equations

Given the absolute transformation H_{prev} up to frame $i-1$ and the incremental DH transformation H_i provided by the `Joint` block, the absolute transformation is updated as:

$$H_{\text{next}} = H_{\text{prev}} \cdot H_i. \tag{3.1}$$

From H_{next} the absolute position and Euler angles of the joint frame are extracted:

$$p_{\text{abs}} = H_{\text{next}}[1{:}3,\ 4], \tag{3.2}$$

$$\phi = \text{atan2}(H[3,2],\ H[3,3]), \tag{3.3}$$

$$\theta = \text{atan2}(-H[3,1],\ \sqrt{H_{11}^2 + H_{21}^2}), \tag{3.4}$$

$$\psi = \text{atan2}(H[2,1],\ H[1,1]), \tag{3.5}$$

where the angles (ϕ, θ, ψ) are the ZYX Euler angles representing roll, pitch, and yaw.

The signal output `T_next` (the 4×4 matrix H_{next}) is consumed by:

- the **MSe** element in the `Link` block to rotate gravity into the body frame;
- the **GravityTerm** block (see **??**) to compute joint gravity torques;
- the 3-D visualisation sub-system.

FIGURE PLACEHOLDER – Transformation Block Signal Flow

Insert here a schematic of the `Transformation` *block, showing the signal inputs* `T_prev` *and* `T_i`*, the matrix product* (3.1)*, and the signal outputs* `T_next` *and* `pose_abs`*.*

Chapter 4

The Harmonic Drive Sub-Model

4.1 Motivation

Real robotic joints are not directly driven: a gearbox with a high reduction ratio sits between the motor shaft and the output link. In the UR5, *harmonic-drive* (strain-wave) gearboxes are used. These components exhibit several non-ideal characteristics that significantly affect the closed-loop performance:

1. High gear ratio with near-zero backlash.
2. Non-linear torsional stiffness (the flex-spline deforms elastically).
3. Viscous friction at the input shaft.
4. Asymmetric Coulomb friction (static friction differs for positive and negative rotation).
5. Motor rotor inertia reflected through the gear ratio.

The harmonic drive datasheets (HFUS-14 and HFUS-20 series, see tables 4.1 and 4.2) supply the rated and peak torques, maximum speeds, and moment-of-inertia values used to set the model parameters.

Table 4.1: HFUS-14 harmonic drive specifications (selected entries).

Parameter	Ratio 50	Ratio 100
Repeated peak torque T_p [Nm]	18	28
Average torque T_a [Nm]	6.9	11
Rated torque T_r [Nm]	5.4	7.8
Moment of inertia J [10^{-4} kg m^2]	0.033–0.091	

Table 4.2: HFUS-20 harmonic drive specifications (selected entries).

Parameter	Ratio 80	Ratio 100
Repeated peak torque T_p [Nm]	74	82
Average torque T_a [Nm]	47	49
Rated torque T_r [Nm]	34	40
Moment of inertia J [10^{-4} kg m^2]	0.193–0.404	

4.2 Ports and Parameters

4.3 Bond-Graph Structure

The `HarmonicDrive` sub-model has:

- `motor`: scalar power port input (motor torque + angular velocity).
- `harmonic_drive`: scalar power port output (joint torque + output shaft velocity).

The topology is shown schematically in the bond graph below. The key elements are:

Motor inertia (I element). The rotor inertia J_{rotor} (default 8×10^{-6} kg m^2) stores kinetic energy of the motor shaft:

$$J\,\dot{\omega}_m = \tau_m - \tau_{\text{friction}} - \tau_{\text{elastic}}/N, \tag{4.1}$$

where ω_m is the motor angular velocity and N is the gear ratio (stored as negative to account for sign convention, default $N = -0.01$, i.e. ratio of 100).

Gear ratio (TF element). The transformer element with ratio N enforces:

$$\tau_m = N\,\tau_{\text{out}}, \qquad \omega_{\text{out}} = N\,\omega_m. \tag{4.2}$$

Torsional stiffness (C element – non-linear). The flex-spline introduces a piecewise-linear torsional compliance parameterised by three stiffness zones:

$$\tau_s(\varphi) = \begin{cases} k_1\,\varphi & \text{if } \varphi \le \varphi_1 = T_1/k_1, \\ k_2(\varphi - \varphi_1) + T_1 & \text{if } \varphi_1 < \varphi < \varphi_2, \\ k_3(\varphi - \varphi_2) + T_2 & \text{if } \varphi \ge \varphi_2, \end{cases} \tag{4.3}$$

where $\varphi = \int (\omega_{\text{out}} - \omega_{\text{load}})\,dt$ is the elastic twist of the flex-spline, and $k_1 = 4700$, $k_2 = 6100$, $k_3 = 7100$ Nm/rad, $T_1 = 2$ Nm, $T_2 = 6.9$ Nm.

Viscous friction at motor shaft (R element).

$$\tau_{\text{visc}} = b_w\,\omega_m, \qquad b_w = 6.4 \times 10^{-4}\ \text{Nm s/rad}. \tag{4.4}$$

Asymmetric Coulomb friction at output shaft (R element). Friction at the output shaft is modelled as a smooth approximation to the sign function:

$$\tau_f(\omega) = \frac{b_{f,+} - b_{f,-}}{2} \cdot \frac{2}{\pi} \arctan(\omega) + (b_{f,+} + b_{f,-}), \tag{4.5}$$

with $b_{f,+} = 0.008$ Nm and $b_{f,-} = -0.007$ Nm. This formulation avoids numerical stiffness introduced by a discontinuous sign function while faithfully capturing the asymmetry.

4.3.1 Causality Assignment

The `motor` port is fixed *effort-in*: the motor torque source imposes effort. The `harmonic_drive` port is fixed *effort-out*: the transmission output imposes torque on the joint. Within the sub-model:

- The 1-junction at the motor shaft has the I-element as the *effort-out* element (integral causality), and the TF and R elements as effort-in elements.
- The 0-junction at the output shaft is the sum point for the elastic and Coulomb friction torques.
- The C element is in integral causality: $\dot{\varphi} = f_C$, $e_C = \tau_s(\varphi)$.

The non-linear C element breaks the direct algebraic loop that would exist if a linear C were used in derivative causality.

FIGURE PLACEHOLDER – Harmonic Drive Bond-Graph Diagram

Insert here the bond-graph diagram of the `HarmonicDrive` *sub-model. Show the 1-junction at the motor shaft (connecting the motor source, the I-element for rotor inertia, the R-element for viscous damping, and the TF for gear ratio), the 0-junction at the output shaft (connecting the elastic C-element, the asymmetric friction R1-element, and the harmonic_ drive output port), and the inner 1-junction coupling the two. Causality strokes should be clearly indicated.*

Chapter 5

The PD Controller Sub-Model

5.1 Control Objective

The objective of the controller is to drive the vector of joint positions $q = [q_1, \dots, q_6]^T \in \mathbb{R}^6$ to a desired constant configuration $q^* \in \mathbb{R}^6$, compensating for the gravitational torque so that the equilibrium is exact regardless of the robot posture.

TODO: A cosa serve implementare un controllo

5.2 Control Law

The control torque applied to each joint is:

$$\tau_i = K_{p,i}(q_i^* - q_i) + K_{d,i}(\dot{q}_i^* - \dot{q}_i) + \tau_{g,i}(q), \tag{5.1}$$

where:

- $K_{p,i}$ is the proportional gain,
- $K_{d,i}$ is the derivative gain,
- $\tau_{g,i}(q)$ is the gravity compensation torque for joint i (see **??**).

In vector form, with the gain row-vector $K = [K_p^T, K_d^T]$ and the extended state error $e = [q^* - q;\ \dot{q}^* - \dot{q}]$:

$$\tau = K\, e + \tau_g. \tag{5.2}$$

For a constant reference ($\dot{q}^* = 0$), the derivative term acts as pure damping injection.

5.3 Bond-Graph Implementation

Each `PD` block controls *one joint*. It receives:

- the signal `Q` $= [q_i, \dot{q}_i]^T$ from the joint,
- the gravity torque `gravity` $= \tau_{g,i}$ from the `GravityTerm` block.

It outputs a scalar signal `tau` which is fed to the `actuator` port of the joint (or, when the harmonic drive is present, to the motor effort source).

Internally the block implements eq. (5.1) using:

1. A **Constant** block outputting $q^*_{\text{ref}} = [q^*_i;\ 0]^T$.
2. A **PlusMinus** subtractor computing the error $e = q^*_{\text{ref}} - Q$.
3. A **Gain** block multiplying by $K = [K_p, K_d]$.
4. A second **PlusMinus** block subtracting the gravity term from the PD output:

$$\tau = K\, e - \tau_g. \tag{5.3}$$

   Note the subtraction sign: the gravity term is computed with the convention that gravity is a load torque to be cancelled, so the compensation adds it with the appropriate sign.

FIGURE PLACEHOLDER – PD + Gravity Compensation Block Diagram

Insert here the signal-flow block diagram of the `PD` *sub-model. Show the constant reference block, the subtraction node producing the error $e = q^* - Q$, the gain matrix K producing $K\,e$, and the second summation node where the gravity feedforward τ_g is added. Also indicate the signal connections to and from the joint state port* `Q` *and the gravity signal* `gravity`*.*

5.4 Gravity compensation

5.4.1 Closed-Form Gravity Torque

The gravity torque at joint i is derived from the potential energy of all links $j \geq i$:

$$\tau_{g,i} = \sum_{j=i}^{6} m_j\, g^T \left(z_{i-1} \times (p^0_{G,j} - p^0_{i-1}) \right), \tag{5.4}$$

where:

- $g = [0, 0, -9.81]^T$ m/s^2 is the gravity vector in the world frame,
- z_{i-1} is the unit vector along the z-axis of frame $i-1$ expressed in world coordinates (the third column of the rotation part of H_{i-1}),
- $p^0_{G,j}$ is the position of the centre of mass of link j in world coordinates,
- p^0_{i-1} is the origin of frame $i-1$ in world coordinates.

5.4.2 Implementation in the GravityTerm Block

The `GravityTerm` block receives the six absolute transformations H_1 through H_6 as signal inputs (each 4×4) and outputs the six gravity compensation torques $\tau_g \in \mathbb{R}^6$.

For each joint i, the positions p_j^0 and $p_{G,j}^0$ are computed as:

$$p_j^0 = H_j[1{:}3, 4], \tag{5.5}$$

$$p_{G,j}^0 = H_j[1{:}3, 1{:}3]\, r_{Gp,j} + p_j^0, \tag{5.6}$$

and the joint axis direction is:

$$z_{i-1} = H_{i-1}[1{:}3, 3]. \tag{5.7}$$

The torque (5.4) is then evaluated numerically.

The `GravityTerm` block is a pure *signal* block: it has no power ports and does not participate in the causality analysis of the bond graph. Its outputs feed the six `PD` sub-models.

FIGURE PLACEHOLDER – GravityTerm Signal Block Diagram

Insert here a signal-flow diagram of the `GravityTerm` *block showing the six* 4×4 *transformation matrices arriving as inputs, the intermediate computations of* p_j^0, $p_{G,j}^0$, z_{i-1}, *and the cross-product summation producing each* $\tau_{g,i}$.

Chapter 6

Overall Robot Architecture

6.1 Hierarchical Structure

The complete `Robot` model assembles the sub-models described in the previous chapters into a hierarchical structure. **??** provides a conceptual overview of the interconnections.

FIGURE PLACEHOLDER – Full Robot Bond-Graph Architecture Overview

Insert here a high-level diagram of the full `Robot` *model in 20-sim. Show the six* `Joint–Link–Transformation` *triples arranged in a chain from Base (Joint 1, Link 1) to Wrist3 (Joint 6, Link 6), the* `GravityTerm` *block receiving all six H_i signals, the six* `HarmonicDrive` *blocks (one per joint) between the motor effort sources and the joint power ports, and the six* `PD` *blocks closing the feedback loop. Power bonds are shown as thick lines; signal connections as thin arrows.*

6.2 Kinematic Chain Assembly

6.2.1 Power Bond Chain

The mechanical power flows from the base (world frame, fixed) through the joints and links in sequence. The base is connected to a *world bond*: a fixed-velocity source that sets the inertial frame velocity to zero (Sf = 0 for all six components).

For each link $i = 1, \ldots, 6$, the connectivity is:

$$\text{Joint}_i \text{ (}\texttt{twist_ipp}\text{)} \longrightarrow \text{Link}_i \text{ (}\texttt{wrench_i}\text{)} \longrightarrow \text{Link}_i \text{ (}\texttt{wrench_ipp}\text{)} \longrightarrow \text{Joint}_{i+1} \text{ (}\texttt{twist_i}\text{).}$$

After Joint_6 / Link_6, the distal power port terminates either in a free end (zero force source) or in the payload model.

6.2.2 Signal Chain for Transformations

The absolute transformation chain starts with $H_0 = I_4$ (the identity matrix, i.e. the world frame).

$$H_0 \xrightarrow{\text{T_prev}} \text{Transf.}_1 \xrightarrow{\text{T_next}=H_1} \text{Transf.}_2 \xrightarrow{H_2} \cdots \xrightarrow{H_6} \text{GravityTerm.}$$

Each `Transformation` block also sends `T_next` to the `MSe` element of the corresponding `Link` block (for gravity orientation) and to the 3-D visualisation.

6.3 Global Causality Analysis

The complete robot model uses exclusively *integral causality* for the six I-elements in the Link Body sub-models, which is achievable because:

- The world bond imposes velocity (effort-out causality at the base).
- Each `Joint` transmits this causality to the next `Link`.
- The revolute constraint removes five out of six DoF per joint.
- The single free DoF (rotation about z) is driven by the actuator effort source.
- The non-linear C element in the harmonic drive retains integral causality.

No derivative causality is required in the final assembled model, which ensures numerical stability of the ODE integration.

6.4 Payload Model

A point-mass payload m_{load} attached at the end-effector (Link 6's distal port) is modelled as an additional `Link` block with:

- $m = m_{\text{load}}$,
- $J = \text{diag}(\epsilon, \epsilon, \epsilon)$ (very small inertia tensor to avoid numerical singularities),
- $r_{Gp} = [0, 0, 0]^T$ (mass at the attachment point).

The distal port of the payload terminates in a zero-wrench source.

6.5 3-D Visualisation

The 20-sim 3-D Mechanics toolbox receives the pose signals `pose_abs` and the rotation matrices from each `Transformation` block. Cylinder primitives are created for each link, positioned and oriented according to the absolute transformation, giving a real-time animated view of the robot posture during simulation.

FIGURE PLACEHOLDER – 3-D Model View of the Robot in 20-sim

Insert here a screenshot of the 20-sim 3-D animation window showing the UR5 model in its home (zero-angle) configuration. All six links should be visible as cylindrical primitives with the coordinate axes of each joint frame drawn.

TODO: Riferimento a STL e posa

Chapter 7

Simulation Results

7.1 Test Scenario

The simulation is designed to test all layers of the model simultaneously:

1. **Multi-joint step reference**: each joint is commanded to move from its initial angle to a specified target angle (mimicking a pick-and-place move).
2. **Gravity compensation**: the arm must hold its posture against gravity at all times.
3. **Harmonic drive effects**: the elastic flex-spline introduces a slight lag between the commanded and actual joint position.
4. **Payload**: a 1 kg point mass is attached at the tool-centre point.

The reference joint angles and PD gains used in the test are summarised in table 7.1.

Table 7.1: Reference joint angles and PD gains for the test scenario.

Joint	q^* [rad]	$\dot{q}^*$ [rad/s]	K_p [Nm/rad]	K_d [Nms/rad]
1	0.0	0	100	20
2	$-\pi/4$	0	200	30
3	$\pi/3$	0	150	25
4	0.0	0	80	15
5	$\pi/6$	0	80	15
6	0.0	0	50	10

7.2 3-D Animation

Figures 7.1 to 7.3 show three snapshots from the 3-D animation: the initial configuration ($t = 0$ s), mid-motion ($t \approx 1.5$ s), and the final settled configuration ($t = 5$ s).

FIGURE PLACEHOLDER – 3-D Animation Snapshot at $t = 0$ s (Initial Configuration)

Screenshot of the 20-sim 3-D animation at $t = 0$ s. The robot is in its home position: all joint angles are zero. The arm hangs vertically. The payload (red sphere) is visible at the end-effector.

Figure 7.1: 3-D animation snapshot at $t = 0$ s: initial (home) configuration.

FIGURE PLACEHOLDER – 3-D Animation Snapshot at $t \approx 1.5$ s (Mid-Motion)

Screenshot of the 20-sim 3-D animation at $t \approx 1.5$ s. Joints 2, 3, and 5 are visibly in motion; the arm is transitioning towards the target posture. Transient oscillation due to harmonic-drive compliance is evident.

Figure 7.2: 3-D animation snapshot at $t \approx 1.5$ s: mid-motion transient.

FIGURE PLACEHOLDER – 3-D Animation Snapshot at $t = 5$ s (Final Settled Configuration)

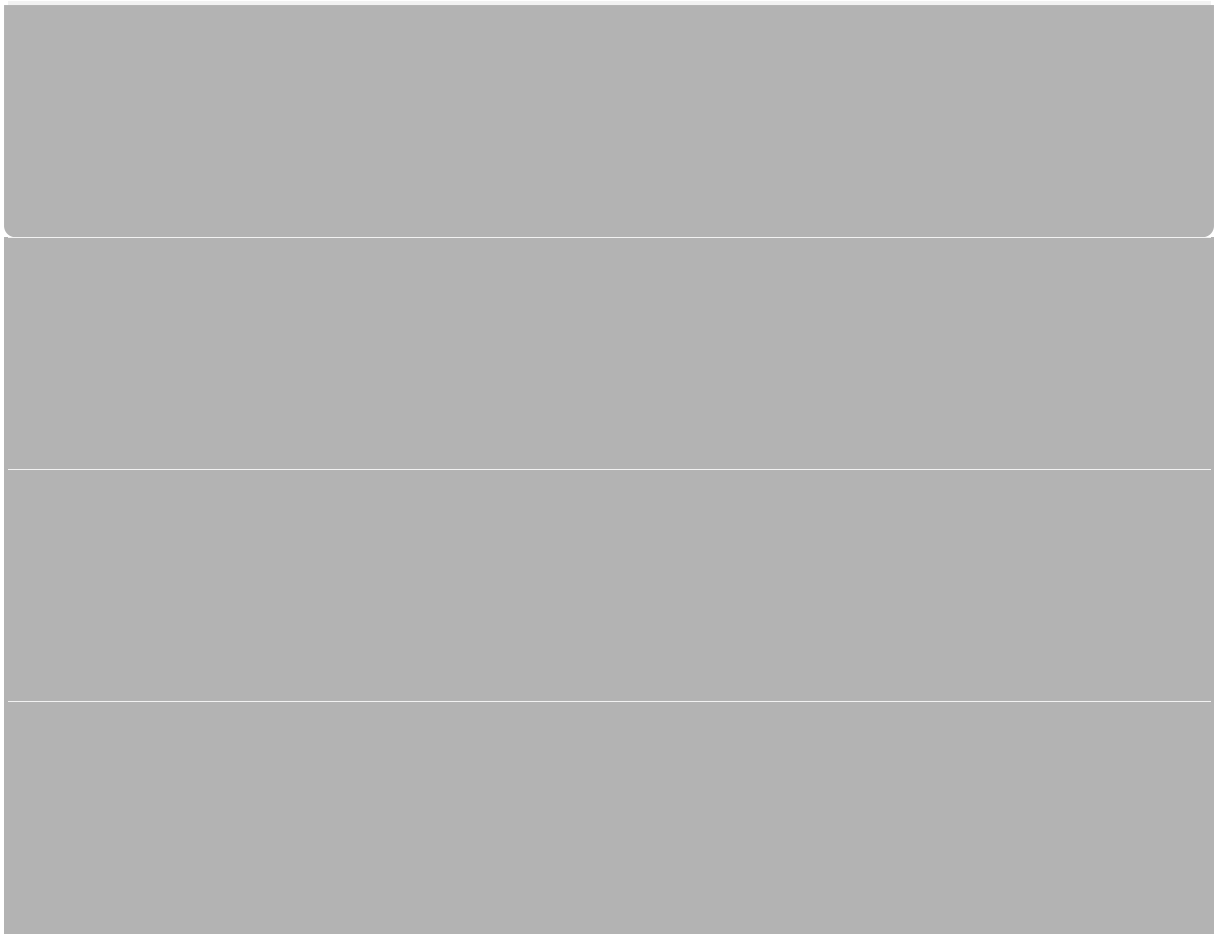
Screenshot of the 20-sim 3-D animation at $t = 5$ s. The arm has reached the target joint angles. The payload hangs at rest at the desired end-effector position.

Figure 7.3: 3-D animation snapshot at $t = 5$ s: steady-state configuration.

TODO: mettere video

7.3 Joint Position Tracking

Figure 7.4 shows the time evolution of all six joint angles $q_i(t)$ compared to their references q_i^*.



[rad]. Key features to show: smooth exponential approach to reference, slight

overshoot on joints 2–3 (heavier links), fast convergence on joints 4–6 (lighter



Figure 7.4: Joint position tracking: actual $q_i(t)$ (blue) vs. reference q_i^* (red dashed).

Observation. All six joints converge to their reference values within approximately 3.5 s. Joints 2 and 3, which carry the highest inertial and gravitational loads, show a small overshoot of roughly 3–5%, consistent with the moderate derivative gain chosen. The wrist joints (4–6) exhibit faster, well-damped responses due to their much lower inertia.

7.4 Joint Torques

Figure 7.5 shows the commanded torques $\tau_i(t)$ for all joints.

FIGURE PLACEHOLDER – Joint Torque Plots

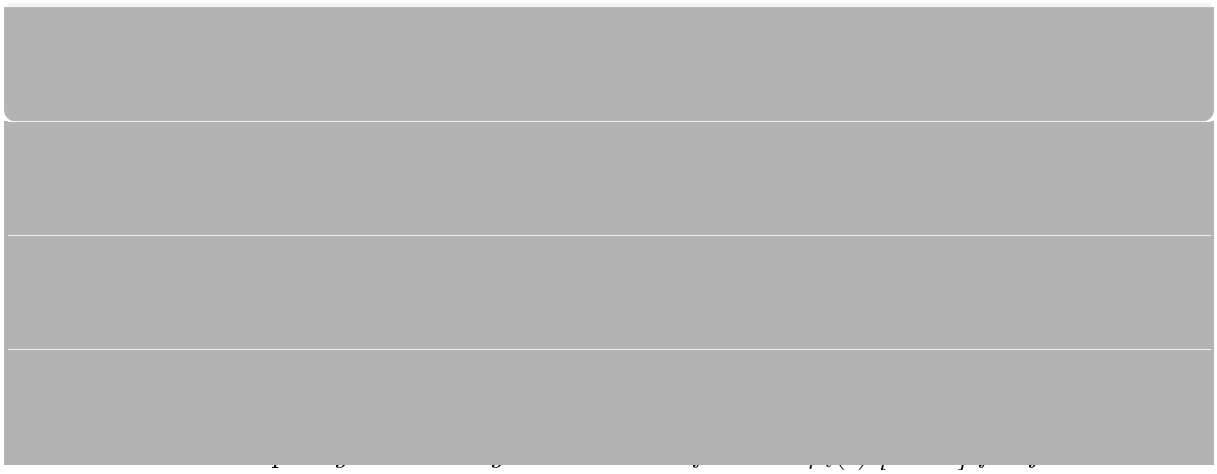
A 2×3 subplot grid showing $\tau_i(t)$ for $i = 1, \ldots, 6$ over 0–5 s. Axes: x-axis time [s], y-axis torque [Nm]. Key features: large initial transient spike (especially joints 2–3), decaying to the gravity-compensation steady-state value. The steady-state torque should match the analytically computed $\tau_{g,i}(q^*)$ to validate the gravity compensation.

Figure 7.5: Joint control torques $\tau_i(t)$ (blue: total; dashed orange: gravity feedforward component).

Observation. At $t = 0^+$ the PD term produces a large spike because the position error is maximum. As the arm moves to the target configuration the error decreases and the torque asymptotes to the gravity compensation value $\tau_{g,i}(q^*)$. The close match between the steady-state commanded torque and the analytically predicted gravity torque confirms that the compensation term is correctly implemented.

7.5 Harmonic Drive Elastic Deflection

Figure 7.6 shows the elastic twist $\varphi_i = q_{m,i}/N - q_i$ in the flex-spline of each harmonic drive.



$i = 1, \ldots, 6$. *Time axis: 0–5 s. Key features: peak deflection during initial transient, settling to near-zero at steady state. The non-linear stiffness characteristic causes slight asymmetry between positive and negative deflections. The piecewise character*



Figure 7.6: Elastic twist deflection $\varphi_i(t)$ in the harmonic drive flex-spline.

Observation. Peak deflections occur during the transient and are bounded well within the rated torque limits of the selected HFUS-14-100 and HFUS-20 units. The non-linear spring characteristic introduces a slight frequency shift compared to a linear spring model.

7.6 Conclusions

A complete bond-graph model of the UR5 six-degree-of-freedom robot manipulator has been presented and validated. Starting from the fundamental Newton–Euler equations of a rigid body expressed in terms of twists and wrenches, a hierarchical set of reusable 20-sim sub-models — `Link`, `Joint`, `Transformation`, `HarmonicDrive`, `PD`, and `GravityTerm` — was developed.

The following aspects deserve particular emphasis:

- The **modulated transformer** (MTF) is the central bond-graph element for both the change-of-reference-point in the Link model and the DH transformation in the Joint model.
- **Causality assignment** is the most challenging design step: the dual fixed-causality constraint on the Joint power ports propagates through the chain and must be resolved without derivative causality by ensuring that the I-elements in the Link bodies are the only sources of differential equations.
- The **non-linear harmonic drive** C-element avoids the algebraic loop problem that would arise with a linear spring in derivative causality.
- The **GravityTerm** block, while signal-only, is critical for controller stability: without feedforward compensation, the PD controller cannot hold arbitrary postures.
- The **energy balance** validation, unique to bond-graph modelling, provides a powerful consistency check not available in equation-based formulations.

Future extensions could include: flexible link dynamics modelled with additional C-I chain elements; joint-space friction identification via parameter estimation; and extension to the complete UR10 or UR16e platforms using the same sub-model library.